

Steve A. Morales

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LANGUAGES AND TECHNOLOGIES

- C++, Python, Java, JavaScript
- Git, GitLab, AWS (EC2), Valgrind, GDB, Vim, Linux, VS Code

EXPERIENCE

Front-End Developer

Tactical Engineering and Analysis

Mar 2025 – Present

San Diego, CA

- Reduced read I/O operations by over 90% by implementing a FileObserver with inotify, verified via iostat
- Developed 100+ GoogleTest unit tests covering over 4 core modules
- Authored 2 widely adopted internal technical guides, reducing onboarding and testing setup time for a 6-engineer team
- Performed memory profiling with Valgrind and gdb to diagnose segmentation faults and memory leaks, enabling downstream fixes
- Configured UDP multicasting over N3N VPN across distributed AWS environments, enabling low-latency data transfer

QA Software Engineer

Programs Management Analytics & Technologies

Jul 2023 – Mar 2025

San Diego, CA

- Authored and standardized C++ test procedures in collaboration with cross-functional teams, improving QA consistency across releases
- Validated C++ code changes through GitLab CI pipelines, improving integration reliability and reducing regression risk
- Configured and customized Docker containers on Linux to create reproducible test environments and improve validation consistency
- Maintained and enhanced C++ build and test workflows on Ubuntu Linux to support consistent release validation

PROJECTS

M.A.R.V.I.N – Medical AR for Surgeons (HoloLens)

Apr 2023 – Jun 2023

- Led development of MARVIN, a medical AR platform using Microsoft HoloLens 2 to visualize 3D organ models and real-time medical data for surgical precision
- Integrated Unity with Microsoft's Mixed Reality Toolkit (MRTK) to build an interactive AR interface supporting hand tracking, voice commands, and spatial mapping
- Converted DICOM-based CT/MRI scans into segmented 3D organ models using open-source ML tools (TotalSegmentator in 3D Slicer), reducing dependency on manual annotation
- Engineered a desktop application using Electron, Node.js, and React for OS-agnostic video input, DICOM upload, and device management
- Implemented video capture and streaming pipeline using WebRTC to relay live ultrasound feed from medical equipment to the HoloLens 2 environment
- Established secure, bidirectional communication using a custom signaling server to support real-time collaboration and remote guidance between surgeons

Autonomous Lane Detection Robot (Jetson Nano)

Jan 2023 – Mar 2023

- Led software integration for an autonomous vehicle project, setting up and configuring the Jetson Nano SBC and microcontroller in a Linux-based development environment
- Utilized ROS (Robot Operating System) for modular robotics programming; developed, sourced, and tested updated nodes and parameters under changing runtime conditions
- Engineered lane detection system using OpenCV and Canny edge filtering to enable real-time computer vision for autonomous driving
- Tuned PID (Proportional-Integral-Derivative) controllers for steering and speed control, improving vehicle stability and lane retention on a dimly lit indoor track
- Dedicated 20+ hours/week to iterative testing and refinement, ultimately achieving successful autonomous lap completion under adverse lighting conditions

EDUCATION

Tempe, AZ	Arizona State University	Aug 2025 – Present
	M.S.E Software Engineering; GPA: 4.0/4.0	
	• Graduate Coursework: Software Verification & Validation Test	
La Jolla, CA	University of California, San Diego	Jun 2019 – Jun 2023
	B.S Computer Engineering; GPA: 3.0/4.0	
	• Undergraduate Coursework: Advanced Data Structures, Design & Analysis of Algorithms, Operating Systems, Computer Architecture, Software Engineering, Computer Vision, Autonomous Vehicles, Intro to Deep Learning, Digital Systems Design, Computer Graphics	

CERTIFICATIONS

CompTIA Security+	Sep 2025
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